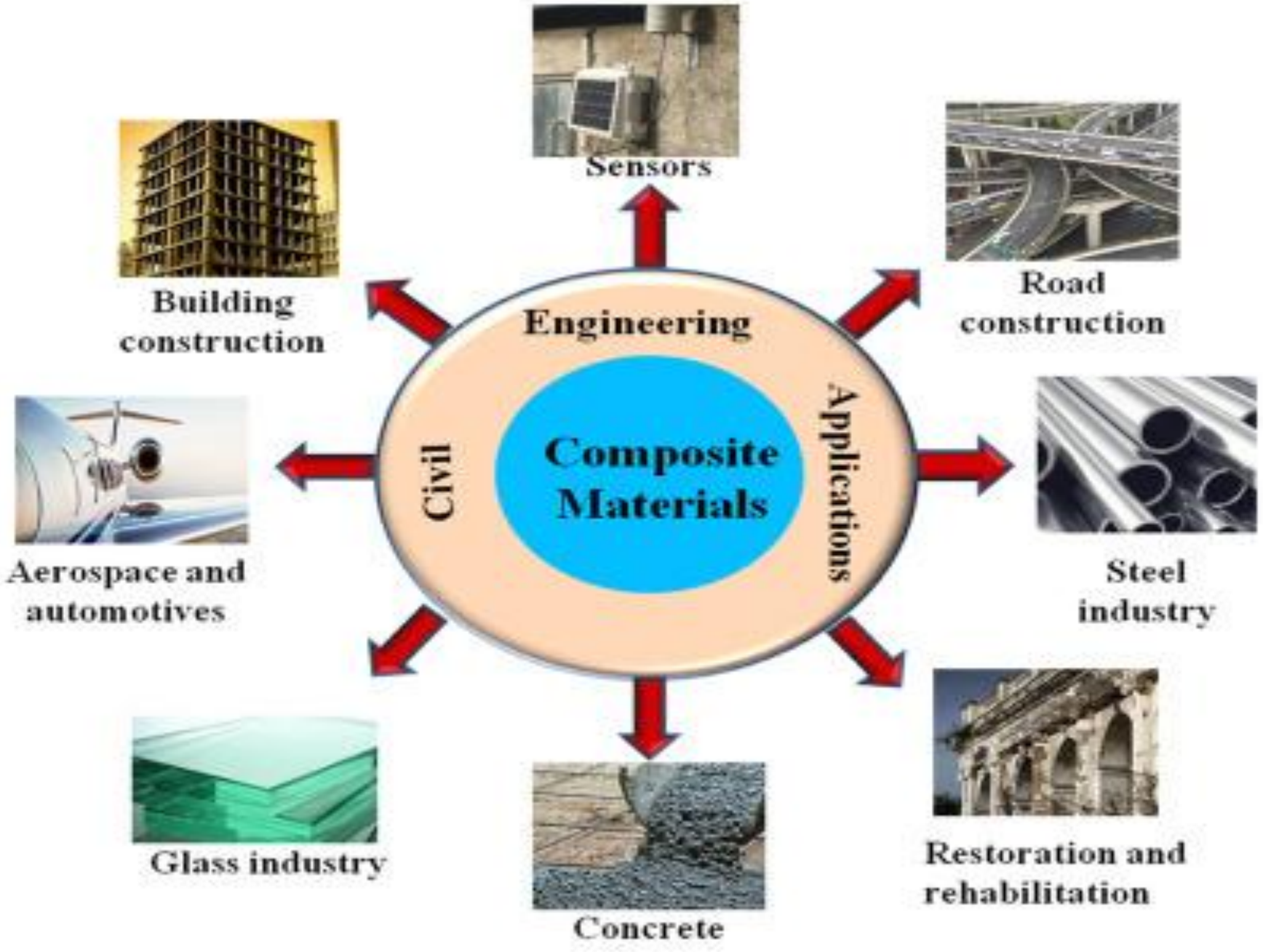
RV College of Engineering®

Unit-V
Engineering polymers and nanocomposites

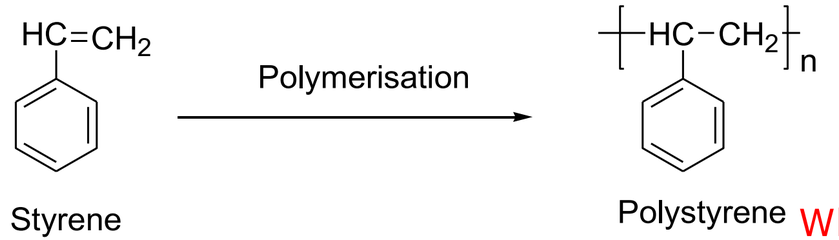
Outline

- Significance of glass transition temperature (Tg) and factors affecting Tg.
- Thermosets-bakelite and epoxy resins.
- thermoplastics- polycarbonate and polyether sulfones- preparation and specific applications in industries.
- Biodegradable polymer: Introduction, synthesis, properties, and application of poly lactic acid (PLA).
- **Reinforcements:** Glass, carbon and natural fibre - synthesis, properties and applications in polymer composites.
- **Testing:** ASTM standards of material testing-tensile strength, flexural strength, ILSS and impact strength.
- Applications of polymer nanocomposites in injection moulded products, paints and 3D printing.


Sensors
Building construction
Road construction
Engineering
Civil
Composite Materials
Applications
Aerospace and automotives
Steel industry
Glass industry
Concrete
Restoration and rehabilitation

- A polymer is a macromolecule having high molecular mass more than 10,000.
- Formation of such molecule requires combination of large number of small molecules. Such small molecules are called **monomers** and they should possess at least two functional groups or unsaturated bonds that can combine with similar or different type of simple molecules.
- The process involved in combining of monomers to get a polymer is called **polymerisation.**
- The number of times the monomer units repeat in a polymer is called **degree of polymerisation.**





Where **n** is called degree of polymerisation.

classification of polymers

Polymers are classified in following ways.

1. Based on method of polymerisation: Addition polymers and Condensation polymers.

2. Based on effect of heat: Thermoplastic and thermosetting polymers

3. Based on the way of repeating units linked: Linear, branched and cross-linked polymers.

4. Based on origin: Natural and Synthetic polymers.

5. Based on arrangement of molecules: Crystalline and amorphous polymers.

Glass Transition temperature

- In crystalline solids (Eg:KCl) there will be orderly arrangement of particles, whereas amorphous solids particles are arranged randomly.
- In case of crystalline solids, a definite quantity of energy is required to separate the particles.
- Hence crystalline solids show sharp melting points. But in case of amorphous solids melting points are not sharp.
- Polymers are generally amorphous in nature as the polymer chains are coiled, interwined and entangled giving a random arrangement.
- Therefore, behaviour of polymers on heating is different from that of crystalline solids.

Glass Transition temperature

- A solid polymer will be hard, stiff and glossy as the chains occupy definite positions in the solid and virtually all the chain movements are frozen.

- When temperature increases the localised units (chain segments) within a long chain molecule starts moving. This movement is called segmental motion.

- If the temperature increases further the whole chain starts moving. This type of movement is called molecular motion.

- **Eg:** 1. The entangled noodles whose ends can be moved but not the whole strand, is segmental motion.
- 2. Earth worm in a bowl that keeps wagging its head and tail without actually moving, is segmental motion.
- The parts of polymer where the local segments have started moving is like liquid phase where as the molecule as a whole whose mobility is forbidden is like solid phase. This state is a combination of solid and liquid state and it is called rubbery state.

Glass Transition temperature

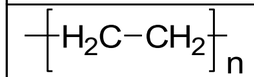
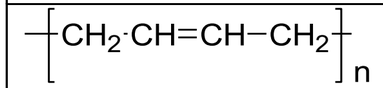
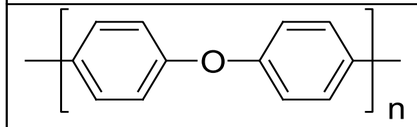
- **The temperature at which a polymer transforms from stiff, hard, glossy state to rubbery state is called glass transition temperature (Tg).**

- The temperature at which the polymer melts and starts flowing is called flow temperature. (Tf or Tm)

Factors affecting Tg

1. Chain Flexibility: Chain flexibility is determined by the ease with which rotation occurs about primary valence bonds.

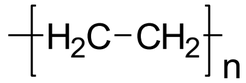
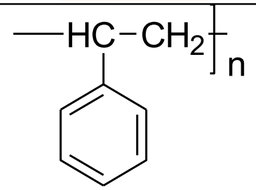
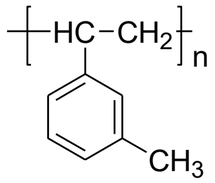
- If there exists a free rotation along carbon chain of the polymer the polymer is flexible.
- Linear polymer chains containing only single bonds have high degree of freedom for rotation and are more flexible.
- More the flexibility more will be the segmental mobility and hence lower will be the Tg.

Polymer	Tg	Remarks
 Polythene	-110°C	Free rotation is possible
 Polybutadiene	-85°C	Contains double bond so there will be restricted rotation
 Polypheneleneoxide	+83°C	Free rotation restricted, not flexible (Rigid)

Factors affecting Tg

2. Presence of side groups: (Bulky groups)

Presence of side groups restricts the free rotation or movement of polymer chain the polymer becomes more rigid and the Tg increases.

Polymer	Tg	Remarks
 Polythene	-110° C	No side groups, free rotation is possible
 Polystyrene	100° C	Presence of phenyl groups on the polymer backbone, becomes rigid, so higher Tg
 Methyl substituted Polystyrene	170° C	Presence of methyl substituted phenyl groups on the polymer backbone, becomes still rigid, so still higher Tg

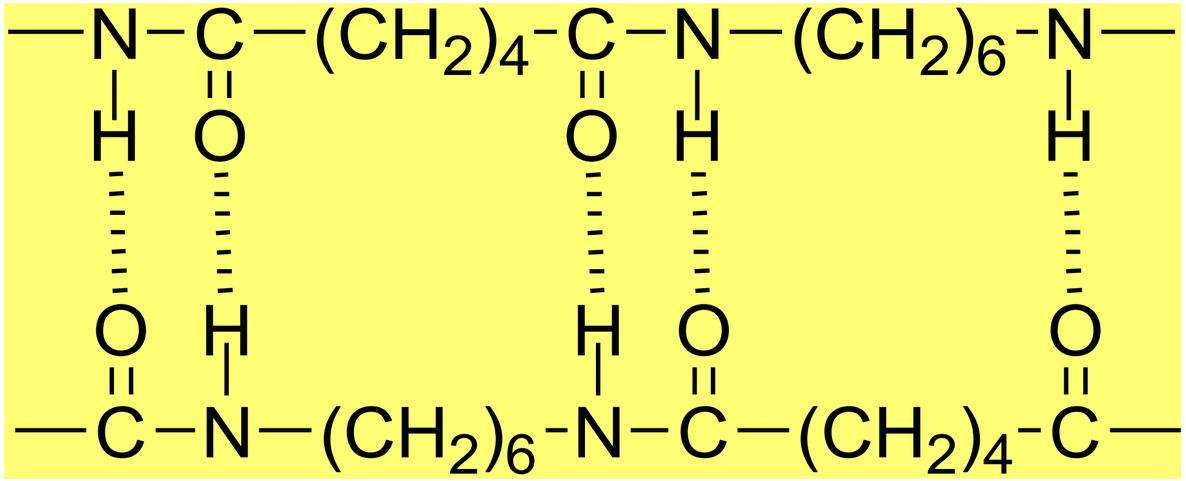
Factors affecting Tg

3. Inter molecular forces:

Presence of intermolecular forces in the polymer chain due to polar groups, dipoles etc exerts strong force of attraction like hydrogen bonding with neighbouring chains.

- This restricts the segmental mobility and hence increases the Tg.

- Eg: Tg of polypropylene is -18° C whereas that of nylon-6,6 is 57° C. This is because nylon-6,6 contains polar amide bonds, there will be hydrogen bonding as shown below.



Factors affecting Tg

4. Molecular mass:

- Tg increases with increase in the molecular mass up to 20,000. Molecular mass beyond 20,000 will have negligible effect on Tg.

- Under high molecular mass, long polymeric chains coil and entangle with one another. This restricts free mobility of the chains and thus increases Tg.

- Under low molecular mass, more number of small chains will have number of loose ends. This leads to segmental motion at the loose end resulting in lower Tg.

Factors affecting Tg

5. Crystallinity:

- Close and orderly arrangement of chains leads to crystallinity in a polymer. When chains are close (as in crystalline solid) intermolecular force of attraction (Vaander Walls force) will be more. So, more energy is required to initiate segmental motion which leads to high Tg.

- Higher crystallinity is expected if the chains are linear without branching.

Factors affecting Tg

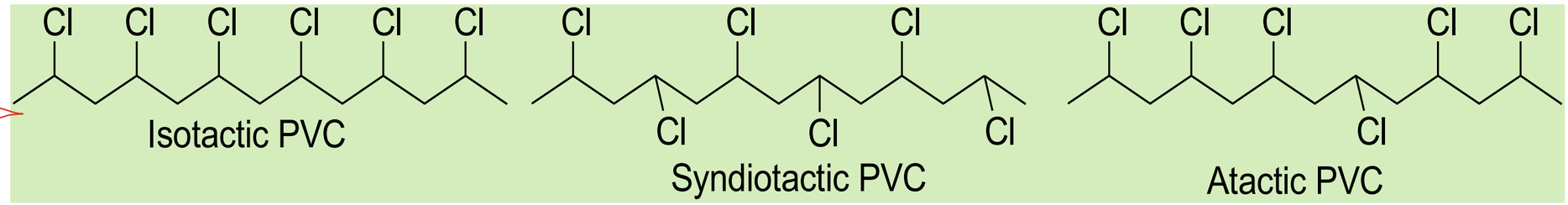
6. Stereoregularity of the polymer:

- Syndiotactic polymer is more symmetric than Isotactic polymer, which is more symmetric than atactic polymer.
- When the structure is symmetric, chains can come closer which leads to increase in crystallinity and thereby increasing Tg of the polymer.
- Thus, Tg of these polymers is in the order of

~~syndiotactic > isotactic > atactic polymer.~~

iso > syndo > atatic //

Eg: Three different confirmations of PVC are as follows.

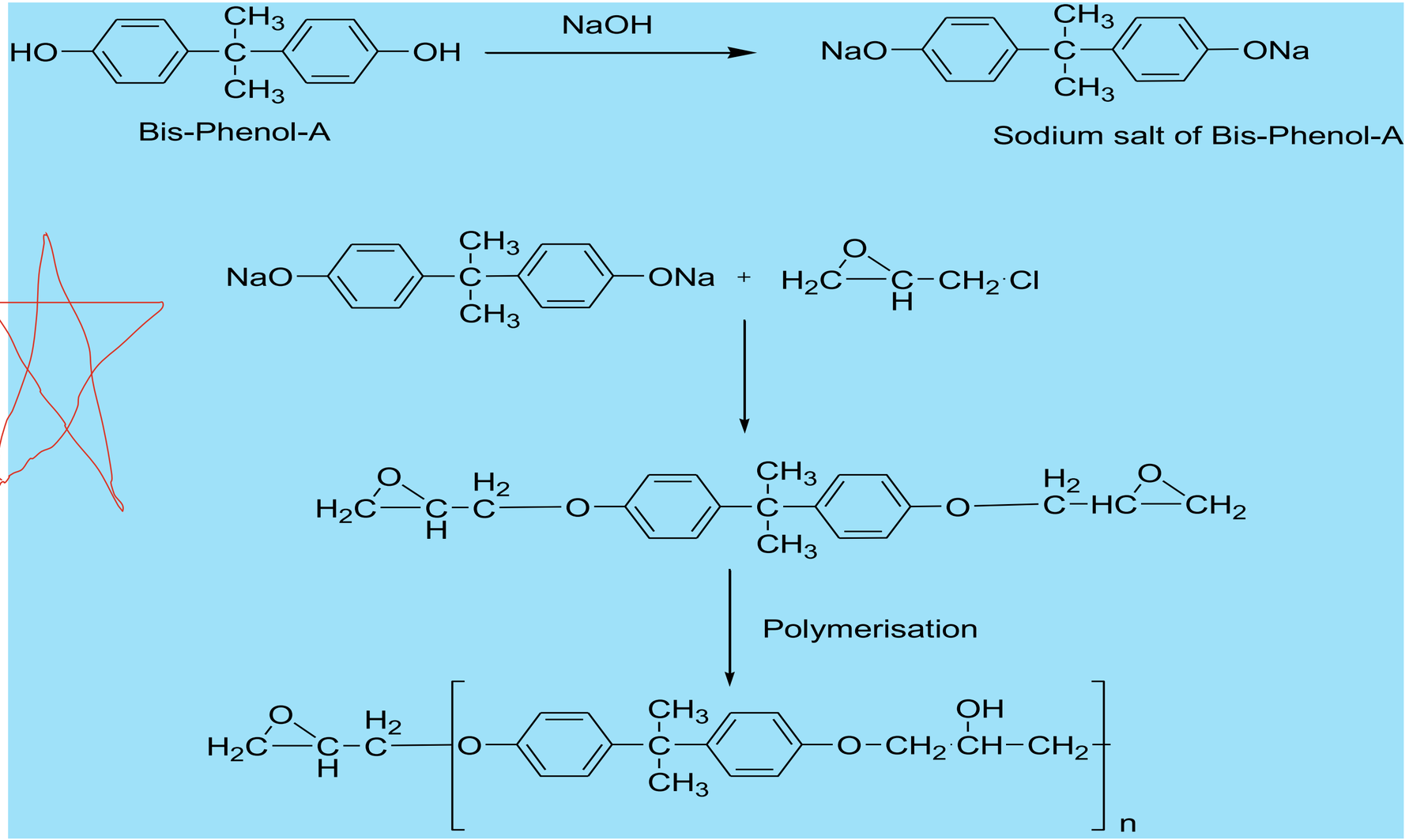


Thermosetting Polymers

- Some polymers undergo chemical change on heating and convert themselves into an infusible mass.
- They are like the yolk of egg, which on heating sets into a mass, and, once set, cannot be reshaped. Such polymers, that become infusible and insoluble mass on heating, are called **'thermosetting"** polymers.
- These polymers are cross linked or heavily branched molecules, which on heating undergo extensive cross linking in moulds and again become infusible.
- These cannot be reused.
- Some common examples are epoxy resins, bakelite, urea-formaldelyde resins, etc.

Epoxy Resins

- These are the polymers contain epoxy group on both the ends.
- The most common epoxy resin is obtained by condensation reaction of epichlorohydrin and Bis-phenol-A in the presence of NaOH as catalyst.



Epoxy Resins

Properties:

1. Epoxy resins possess good electrical insulating properties.

2. Epoxy resins are resistant to water, acids, alkalis and various organic solvents.

3. They offer very good wearing and abrasion resistance.

4. The polar nature of the resins makes them excellent adhesives.

5. Cross linked polymers have high toughness and heat resistant properties.

Epoxy resins: (Araldite)

Applications:

1. It is used as an adhesive to bind wood, glass, concrete, ceramic, metallic and leather materials.
2. Epoxy resins are applied over cotton, rayon and bleached fabrics to impart grease resistance and shrinkage control.
3. Used in manufacture of skid resistant industrial floorings and highway surfacing.
4. Used as laminating materials for electrical equipments.
5. Moulds which are made from epoxy resins are used in the production of aircraft and automobile components.

Note: Curing of epoxy resins

Epoxy resins are converted in to three dimensional cross linked polymers by reaction with curing agents such as polyamides, polyamines, dicarboxylic acids, acid anhydrides, etc. The process is called curing of epoxy resins. On curing epoxy resins show outstanding properties of toughfness, chemical inertness, flexibility and strong adhesion.

EPOXY RESINS: (ARALDITE)

Applications:

1. It is used as an adhesive to bind wood, glass, concrete, ceramic, metallic and leather materials



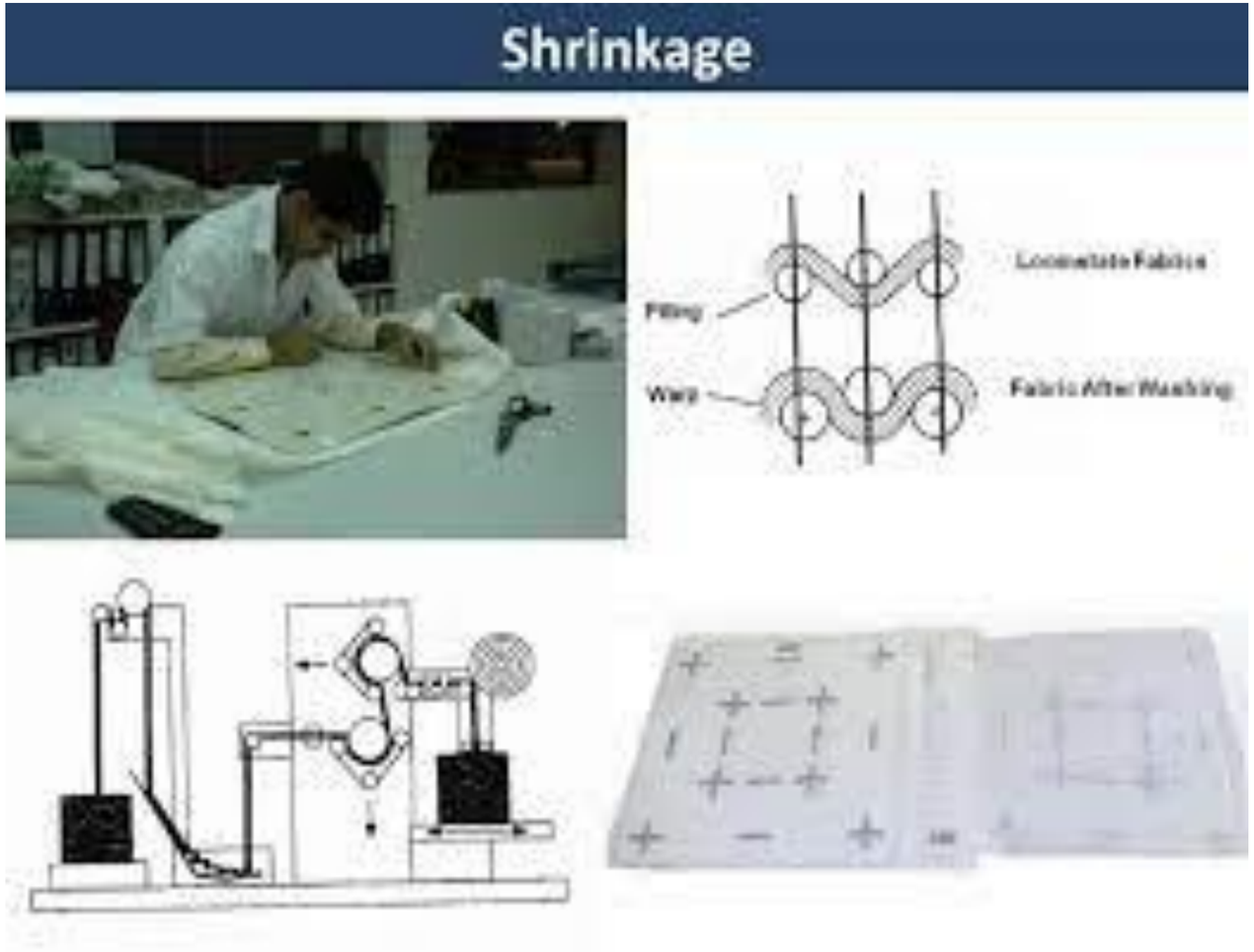


EPOXY RESINS: (ARALDITE)

Applications:

2.Epoxy resins are applied over cotton, rayon and bleached fabrics to impart grease resistance and shrinkage control.





EPOXY RESINS: (ARALDITE)

Applications:

3.Used in manufacture of skid resistant industrial floorings and highway surfacing.



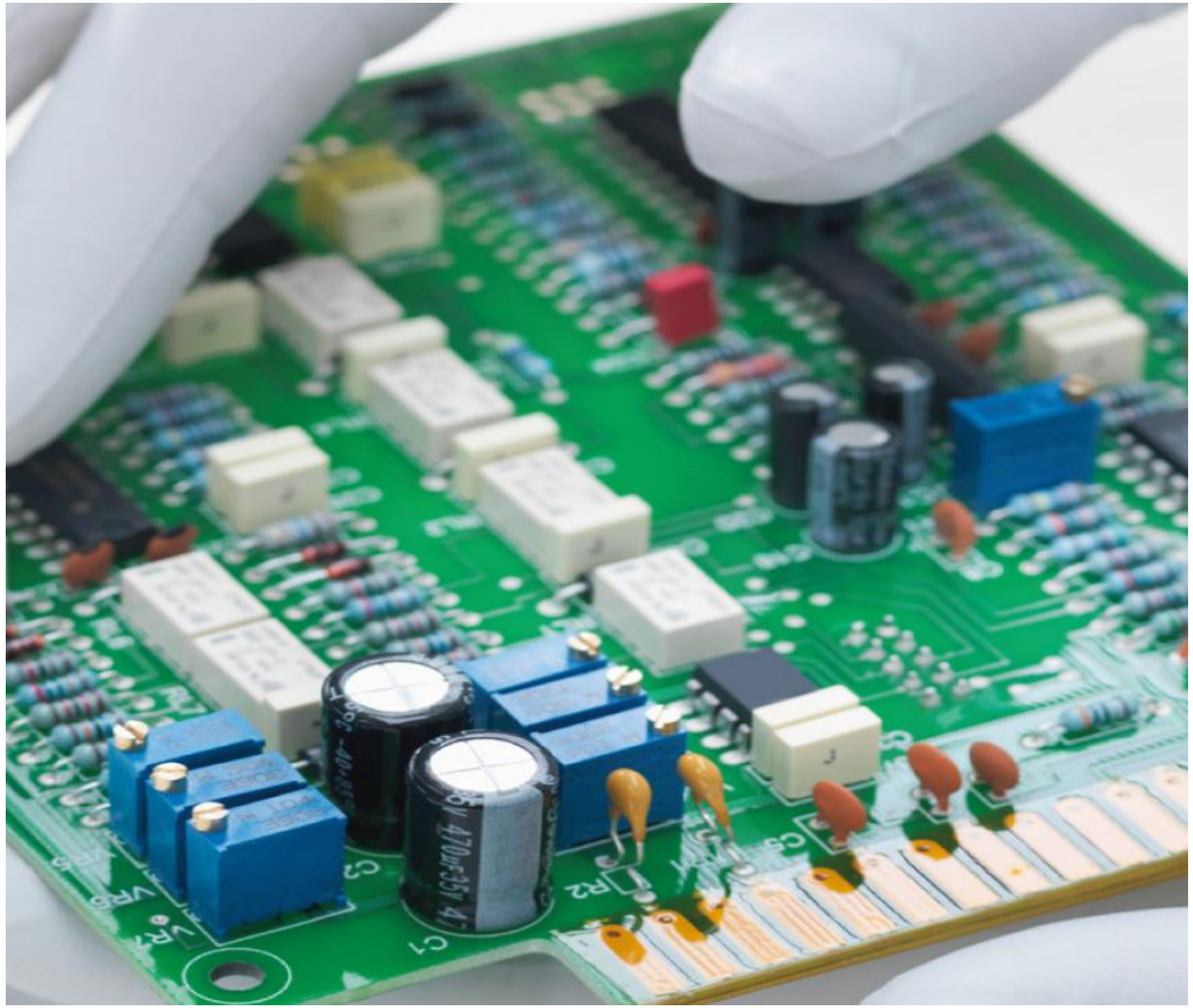




EPOXY RESINS: (ARALDITE)

Applications:

4. Used as laminating materials for electronics/ electrical equipment's.



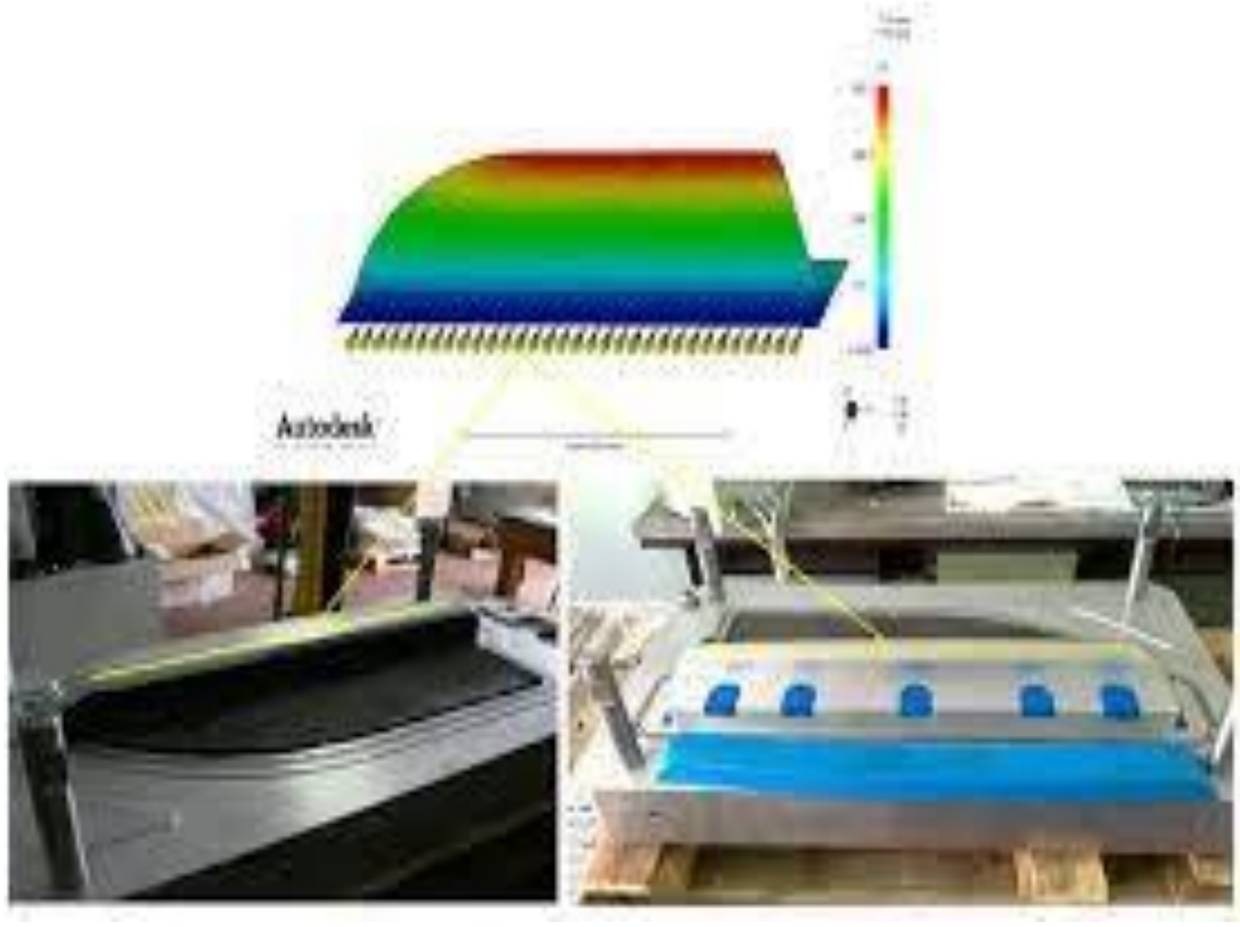


EPOXY RESINS: (ARALDITE)

Applications:

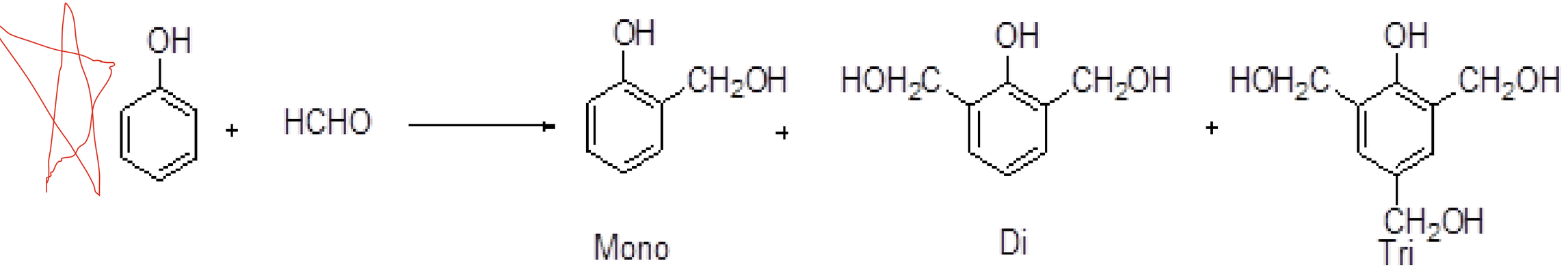
5.Moulds which are made from epoxy resins are used in the production of aircraft and automobile components.





Bakelite (Phenol formaldehyde resins)

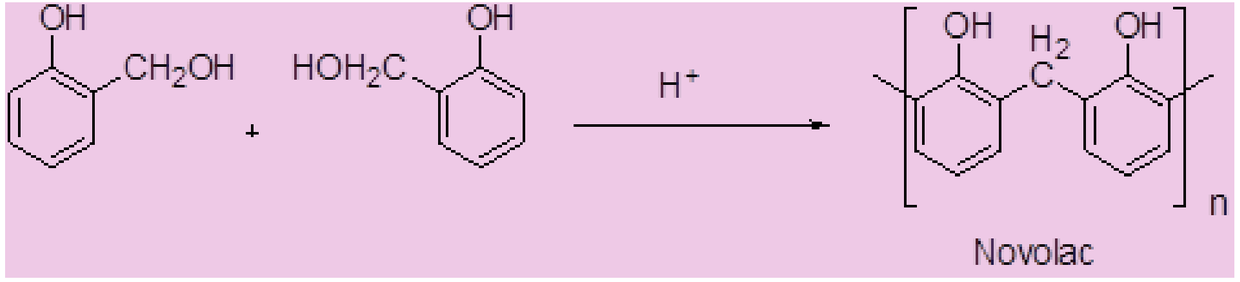
- Phenol-Formaldehyde (PF) resins are formed by the poly condensation between phenol and formaldehyde.
- The polycondensation reaction can be catalyzed either acids or bases.
- Aqueous solution of formaldehyde with phenol is heated to 70-75 C. The monomers like monomethylol phenol, dimethylol phenol, trimethylol phenol are formed.



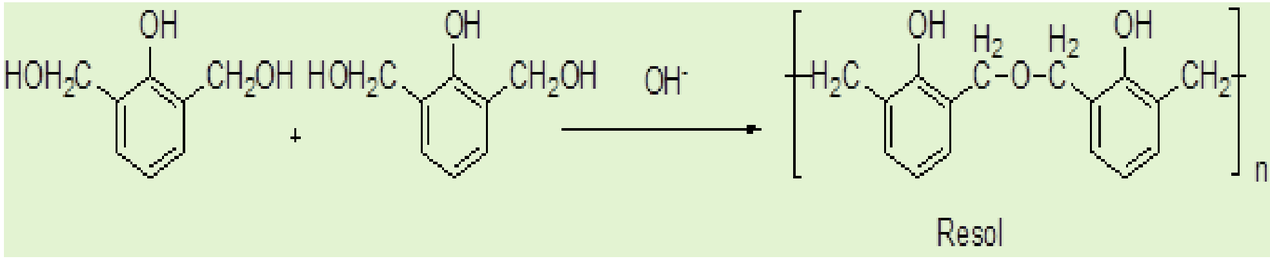
Bakelite (Phenol formaldehyde resins)

The monomers condense to give a polymer. The nature of the reaction depends on the ratio of phenol to formaldehyde in the solution.

Case (i) If the phenol to formaldehyde ratio is greater than one (P/F>1), the monomethylol phenol is expected in higher percentage. This monomethylol phenol undergoes polymerisation in presence of acidic catalyst forms a polymer called Novolac.



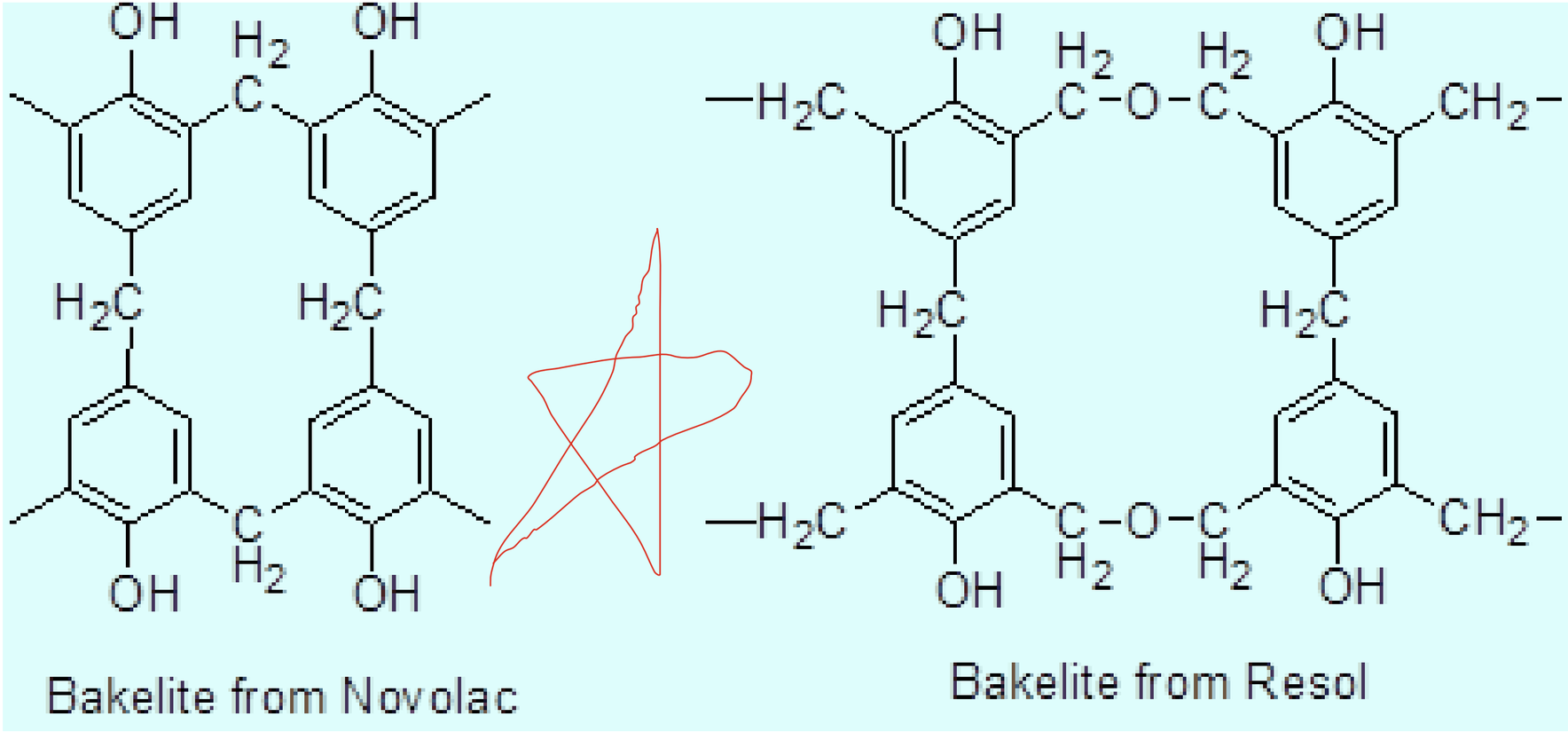
Case(ii) If the phenol to formaldehyde ratio is less than one (P/F<1), the di and trimethylol phenol are expected in higher percentage. This di and trimethylol phenols undergo polymerisation in presence of base catalyst forms a polymer called Resol.



Bakelite (Phenol formaldehyde resins)

- Novolac or resol undergo curing when mixed with hexamine (Hexamethylene tetramine).

- Hexamine helps in the cross linking of the chains. The highly cross-linked product is called Bakelite.



Bakelite (Phenol formaldehyde resins)

Note:

1. Novolac or resol resins when mixed with drying oil are used as varnishes.
2. Both novolac and resol are linear polymers therefore thermoplastic in nature.
3. Resol is characterised by ether linkage whereas novolac is characterised by methylene linkage.
4. The curing agent, hexamine acts as a source of formaldehyde when it reacts with water.

Properties of phenol-Formaldehyde resins:

1. Bakelite is resistant to water, acids, and organic solvents.
2. Due to the presence of phenolic –OH group it is attacked by alkalies.
3. Good electrical insulator and got excellent adhesive properties.
4. Rigid, hard and scratch resistant thermosetting polymer.

Bakelite (Phenol formaldehyde resins)

APPLICATIONS:

1. Used in the manufacture of heater handles, TV and radio cabinets, because of it high temperature resistance.







Bakelite (Phenol formaldehyde resins)

2. As Bakelite is good insulator, it is used for making electric insulator parts like switches, plugs, and switch boards.







Bakelite (Phenol formaldehyde resins)

3. For making brake linings, abrasive wheels.







Bakelite (Phenol formaldehyde resins)

4. To seal electrical bulbs to their caps.









Bakelite (Phenol formaldehyde resins)

5. For making decorative laminates and wall papers.









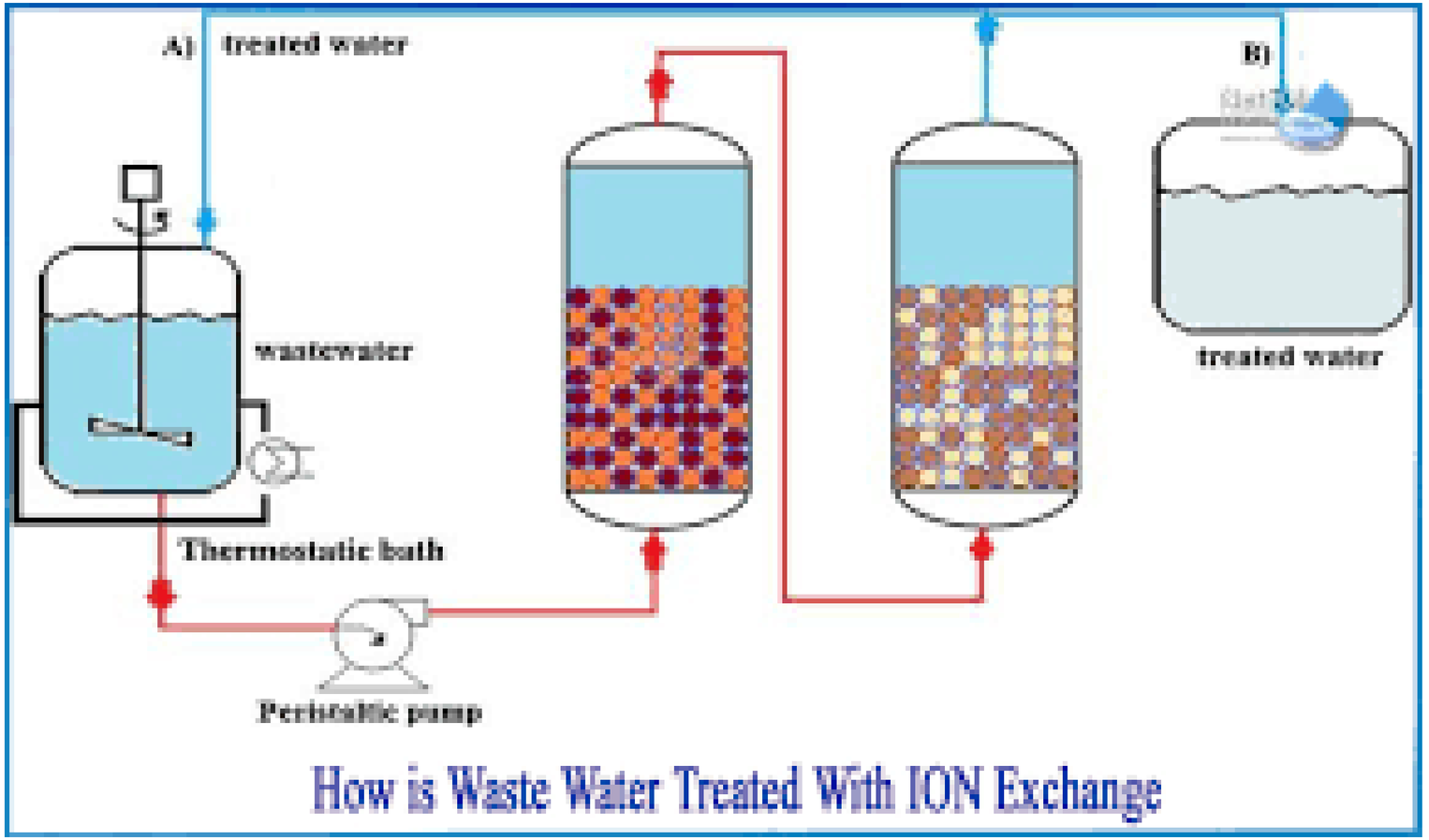




Bakelite (Phenol formaldehyde resins)

6. In the production of ion exchange resins used in the purification of water.





THERMO PLASTIC POLYMERS

- Polymers soften on heating and can be converted into any shape that they can retain on cooling.
- These are the **linear or slightly branched long chain** molecules.
- Polyethylene, PVC, nylon and sealing wax are examples of thermoplastic polymers.

Poly carbonate

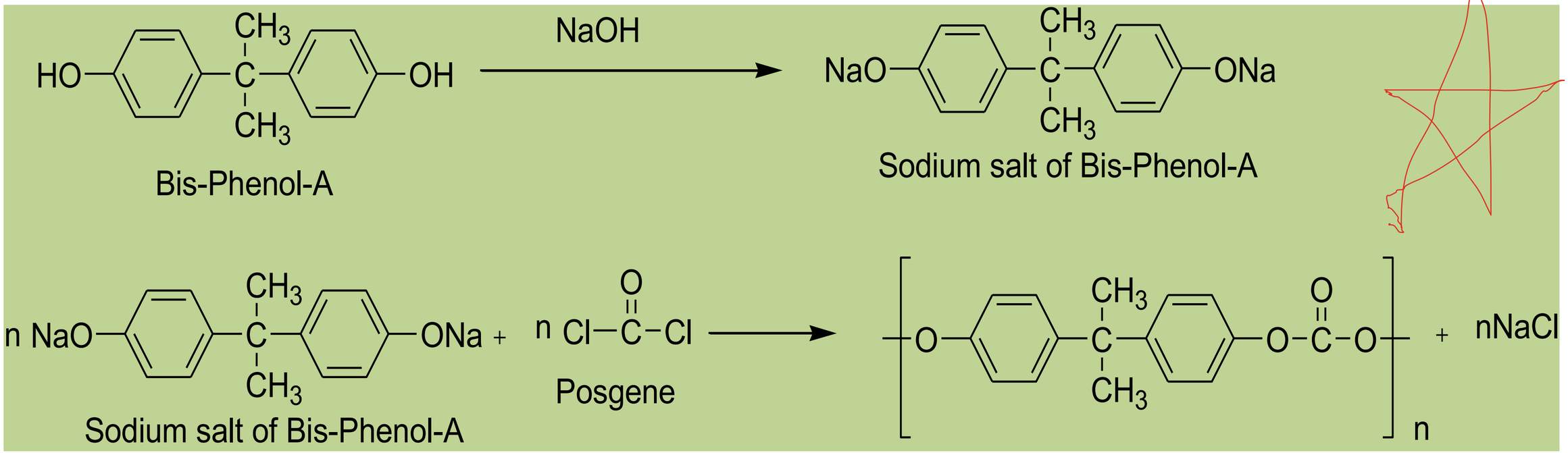
Polycarbonates are the polyesters of unstable carboxylic acids. They contain [-O-CO-O-] linkage through the chain.

- Polycarbonates are prepared by condensation of

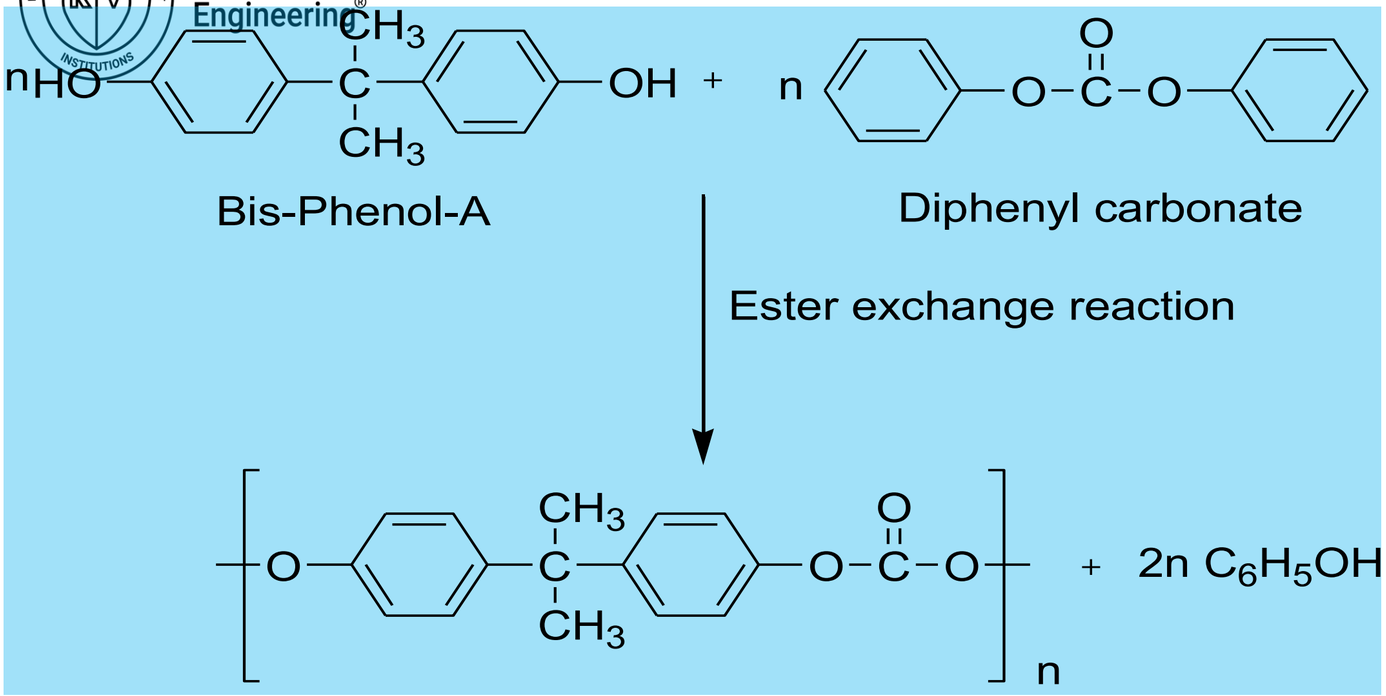
(i) Posgene and sodium salt of Bis-phenol-A

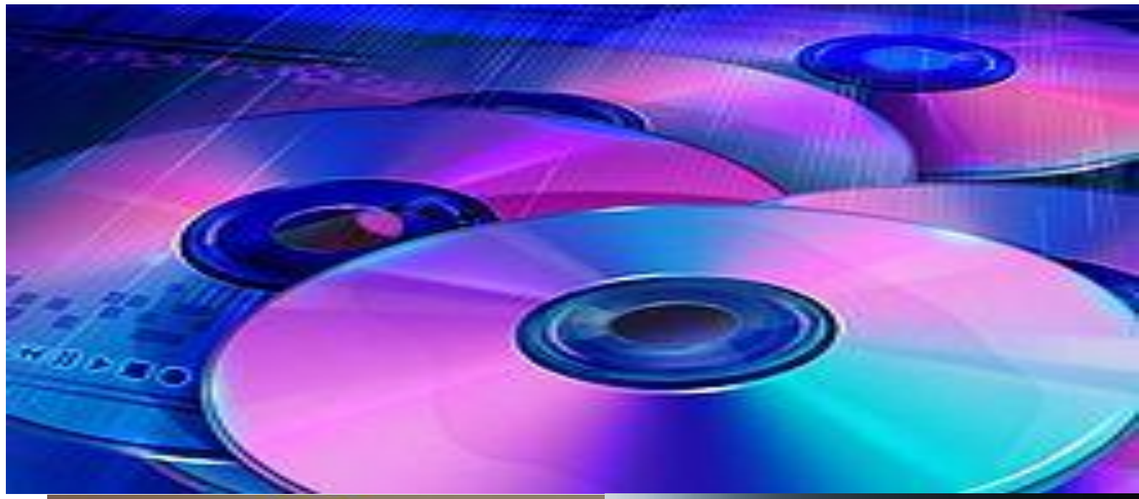
(ii) Diphenyl carbonate and Bis-phenol-A

Method(i): From Posgene and sodium salt of Bis-phenol-A



Method (ii): From Diphenyl carbonate and Bis-phenol-A









Properties:

1. It has high tensile strength and impact resistance.
2. It is white transparent thermoplastic polymer.
3. Highly soluble in alkalies and acids
4. Got very high melting point.

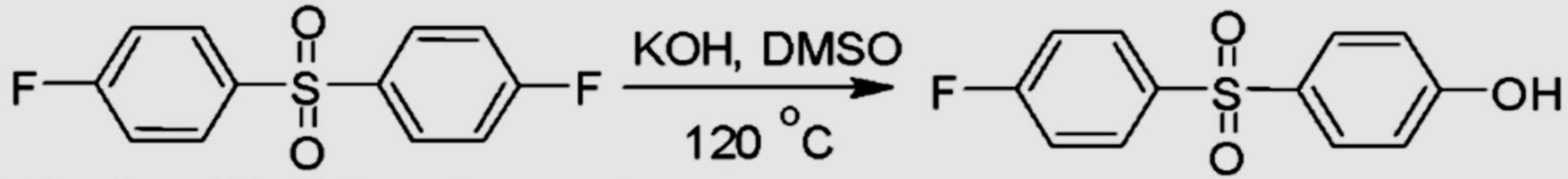
Applications:

1. Used in the manufacture of many useful articles such as telephone parts, unbreackable glazing appliences, etc due to its high tensile strength.
2. Used in the manufacture of safety goggles, automobile lenses, DVD, CD etc due to its transparent nature.

Poly ether sulfone

Synthesis:

Step 1

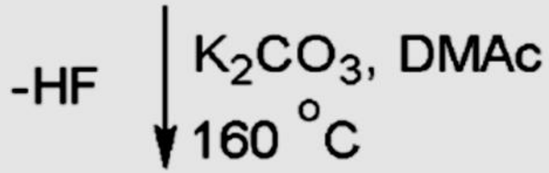


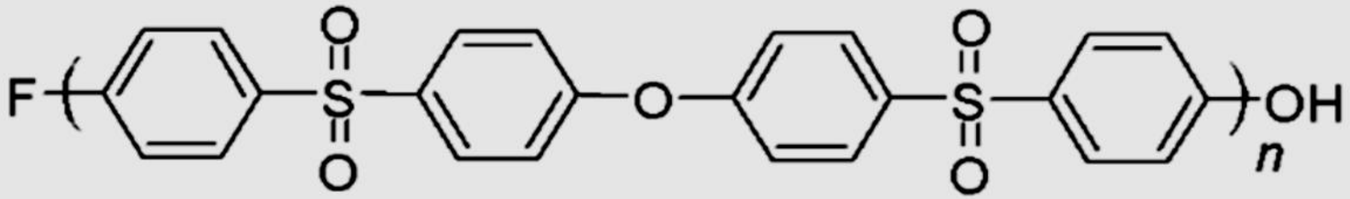
1,1'-sulfonylbis(4-fluorobenzene)

4-(4-fluorobenzene-1-sulfonyl)phenol

Step 2







Poly ether sulfones

Poly ether sulfone

Properties:

Due to rigid structure of poly sulfone, they are very stable and exhibit excellent properties such as,

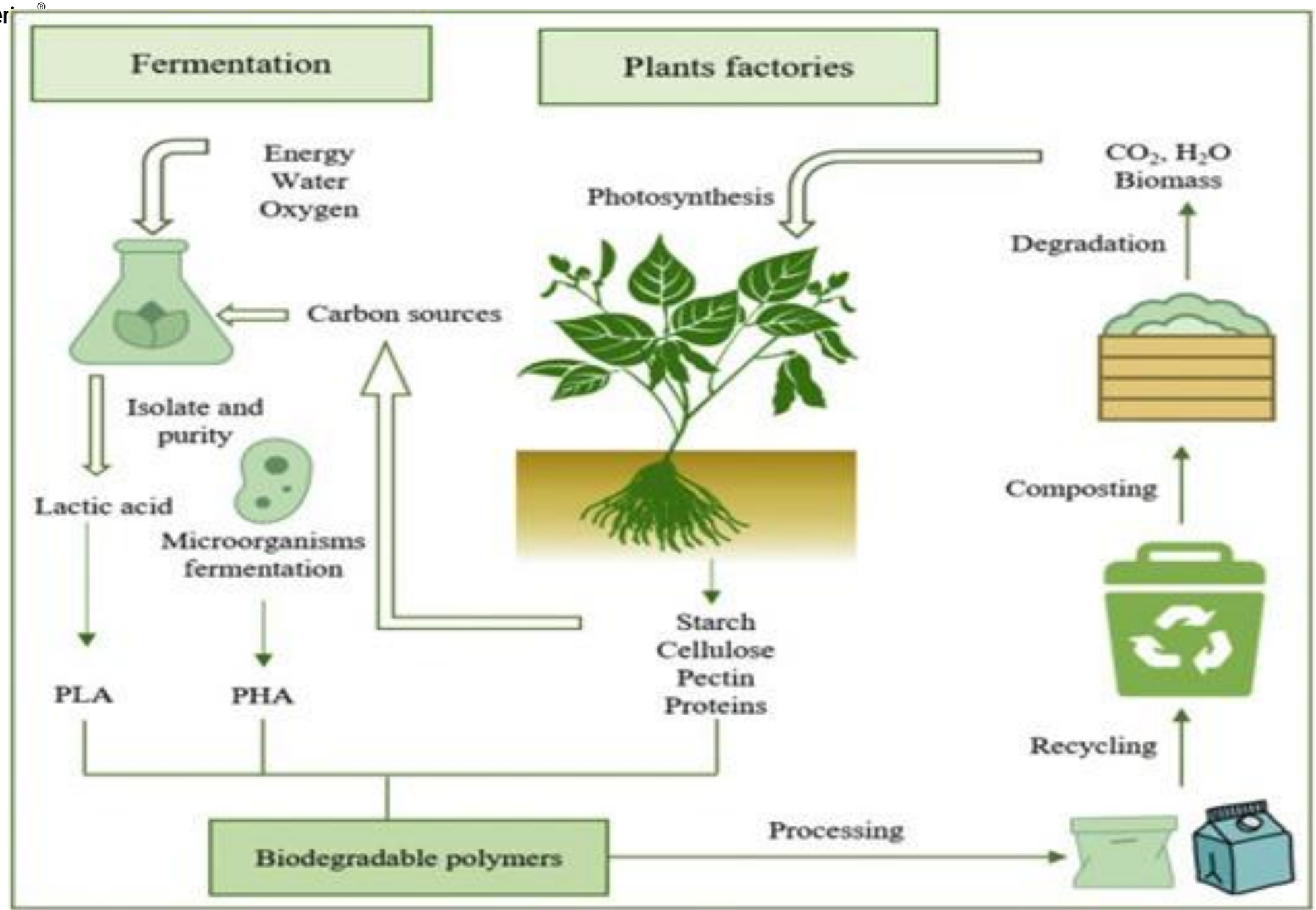
- Excellent heat resistance
- Hydrolysis resistance to aqueous and alkaline media
- High mechanical strength with excellent young’s and rigidity modulus
- Highly transparent and excellent flame retardancy
- The glass transition temperature ranges from 190-230C
- High dimensional stability

Poly ether sulfone

Applications:

- In the preparation of membrane for water purifications
- For gas and chemical separations using membrane technology
- Used as composite materials along with glass fibres in mechanical applications
- As membrane in proton exchange membrane fuel cells
- As food storage containers because of their thermal stability
- In electrical equipment’s
- Because of their thermal stability it is extensively used in aerospace applications
- As heat retardant materials in automobiles

BIODEGRADABLE POLYMERS



BIODEGRADABLE POLYMERS

Plastic bucket kept in sunlight and rain for long time loses its lustre and strength.

- This deterioration in properties is due to a phenomenon called 'polymer degradation', which is characterised by an uncontrolled change in the molecular weight or constitution of the polymer.
- The degradation is a reduction in the molecular weight of the polymer.
- "Thus the biodegradable polymers are the polymers which will degrade by the action of naturally occurring microorganisms like bacteria, fungi or sunlight".

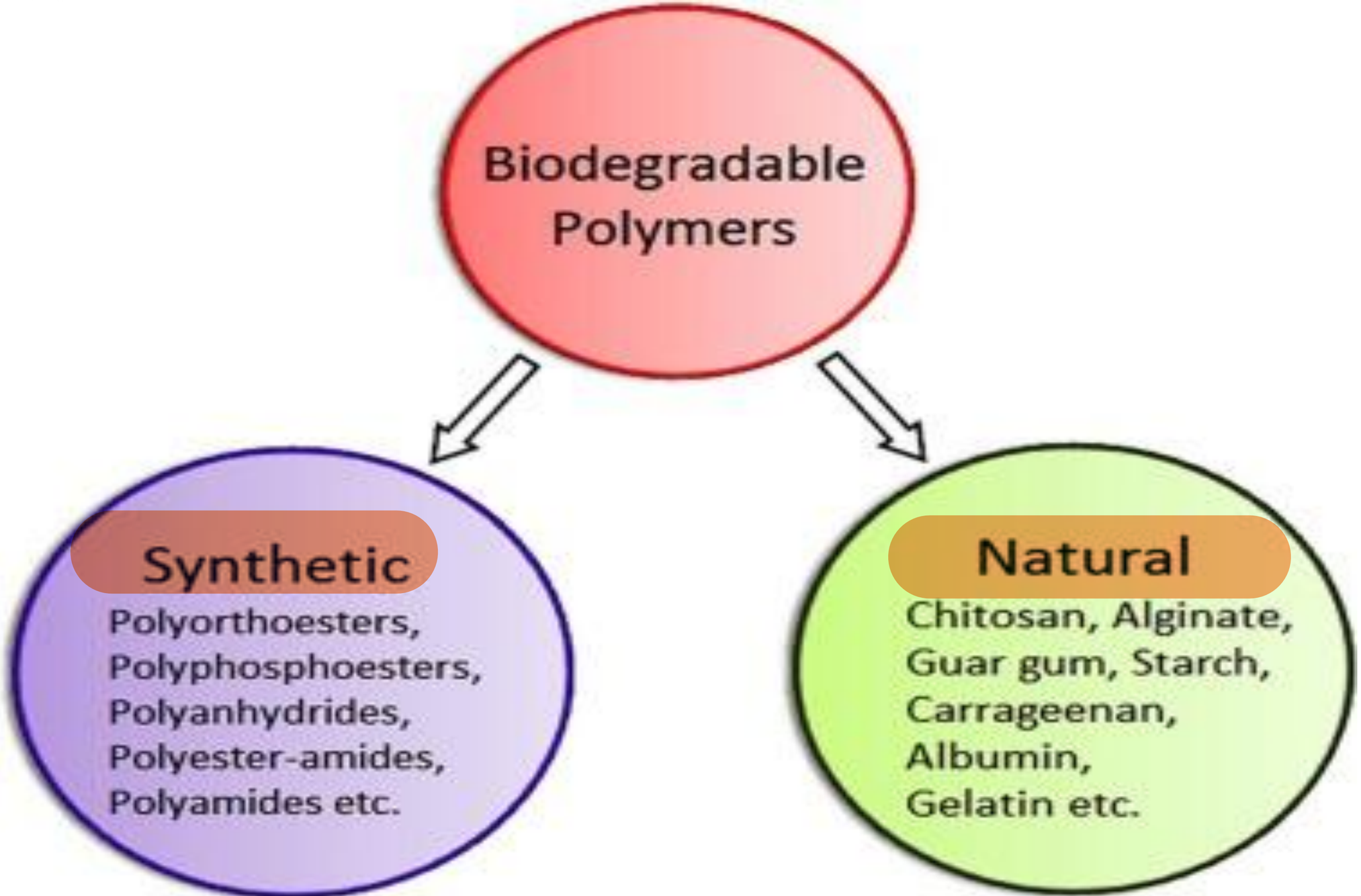
Requirements of Biodegradable polymers:

1. Biodegradable polymers should have hydrolysable linkages like esters, amides or ether.

2. They should be hygroscopic in nature.

3. The product formed after degradation should act as compost.

BIODEGRADABLE POLYMERS



Poly Lactic acid (PLA)

Properties:

- The PLA is a semi-crystalline polymer with glass transition temperature around 55 to 59°C and melting point 174-184 ° C.
- It shows a good mechanical strength, high Young's modulus, thermal plasticity and has good processability.
- It is unstable in wet conditions, which can undergo chain disruption in the human body and degrades into nontoxic by - products, lactic acid, carbon dioxide and water which are subsequently eliminated through the Krebs cycle and in the urine.

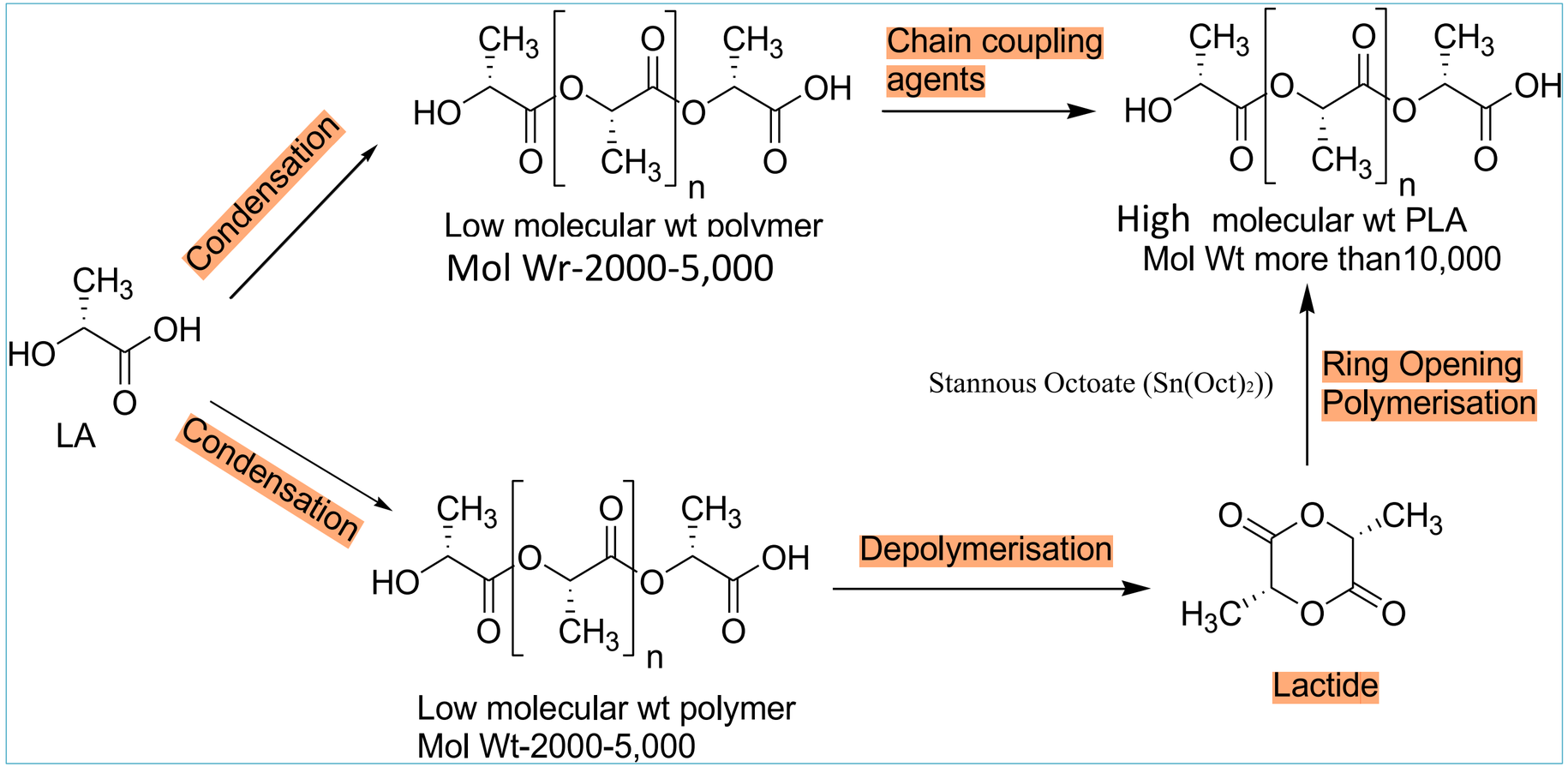
Synthesis of Poly Lactic Acid:

There are two important methods for PLA synthesis:

(i) Direct polycondensation of lactic acid and
(ii) Ring opening polymerization of lactic acid cyclic dimer, known as lactide.

Poly Lactic acid (PLA)

Following reactions show the reaction mechanism.



Note

- In direct condensation, solvent is used and higher reaction times are required.
- Ring-opening polymerization (ROP) of the lactide needs catalyst (Stannous octoate ($Sn(Oct)_2$)) but results in PLA with controlled molecular weight.

Applications of biodegradable polymers

- Polycaprolactum can be used as micropsheres for drug delivery systems.
- Polycaprolactum can be used as medical implants.
- Ultrahigh-strength PLA are used to make bone nails and screws

PLA are used in Fracture fixation and Ligament augmentation

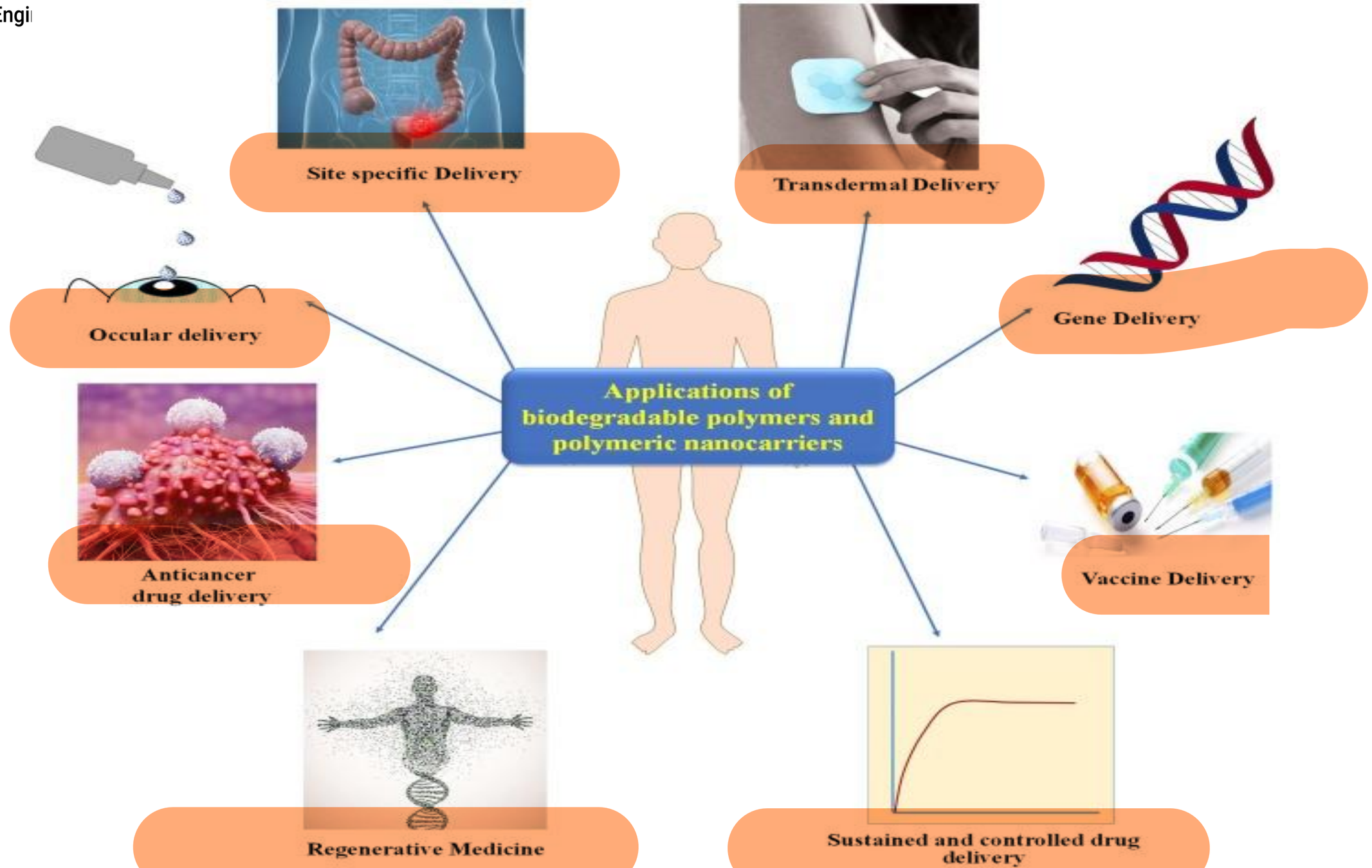
Medical Applications

Wound management
Sutures
Staples
Clips
Adhesives
Surgical meshes

Orthopedic devices

Pins
Rods
Screws
Tacks
Ligaments

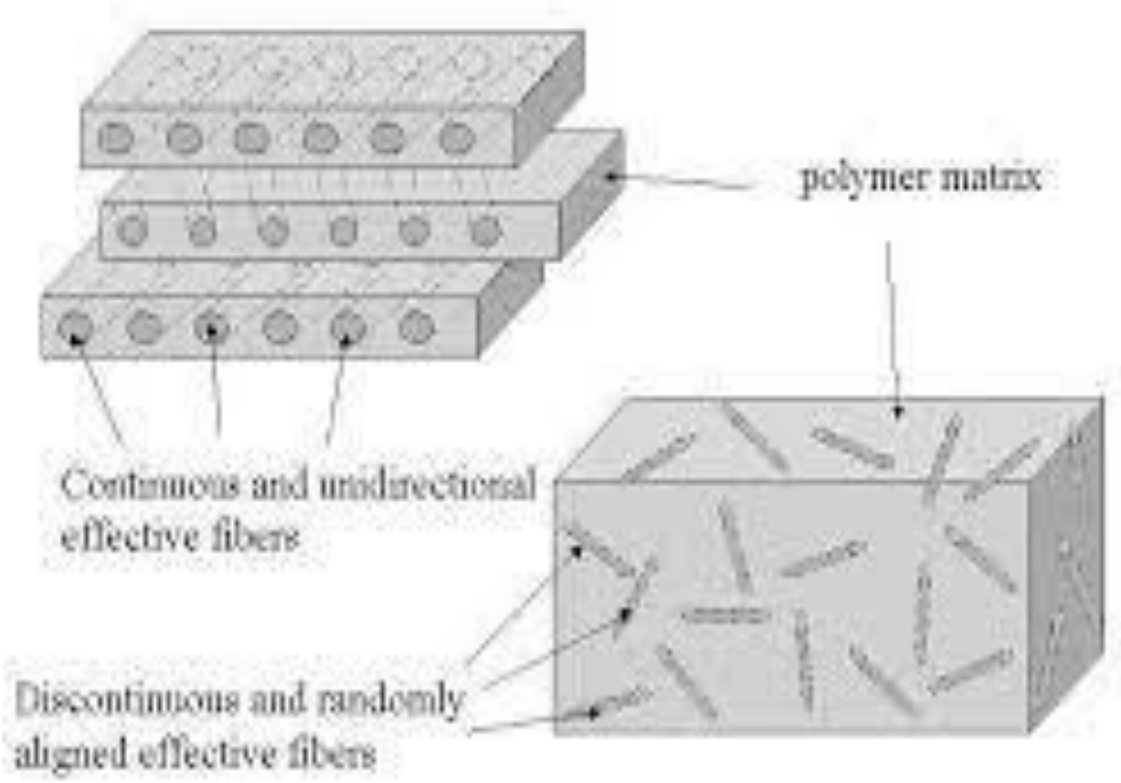
Applications of biodegradable polymers

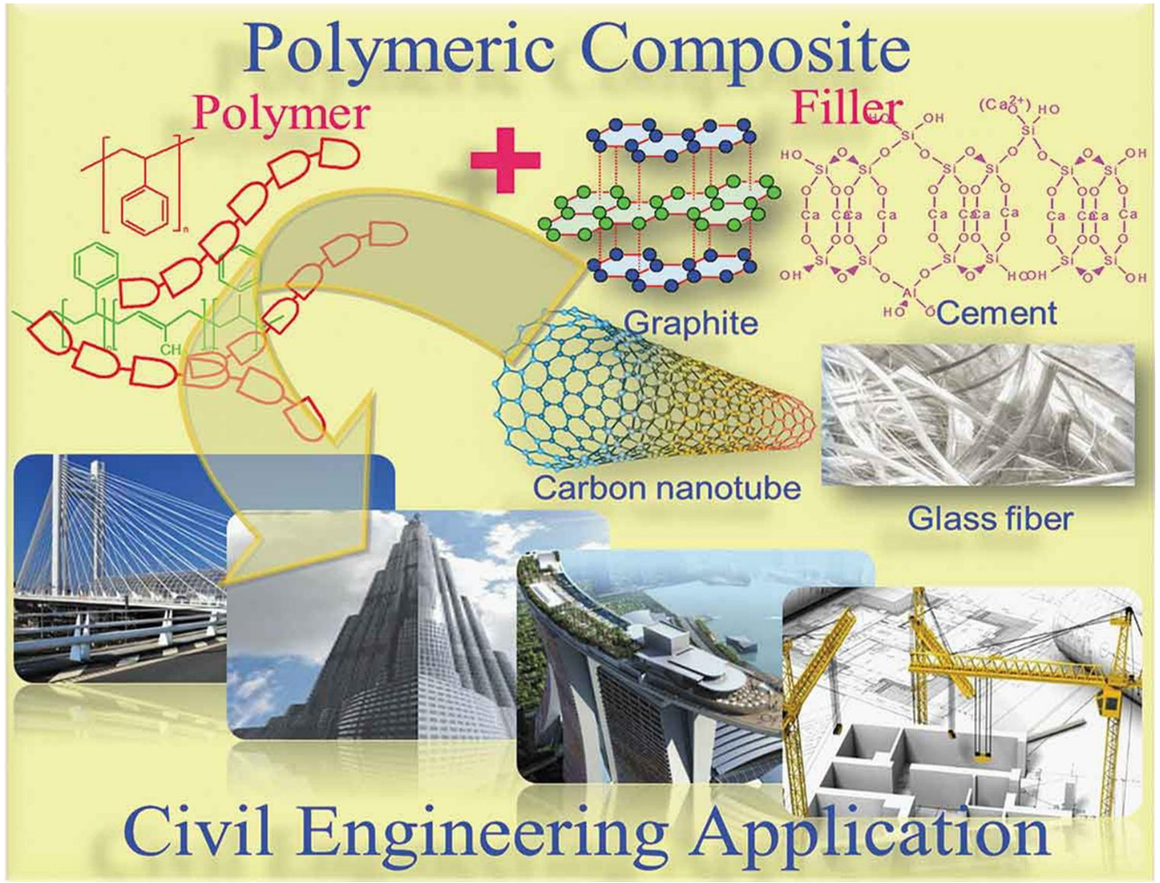


Polymer Composites

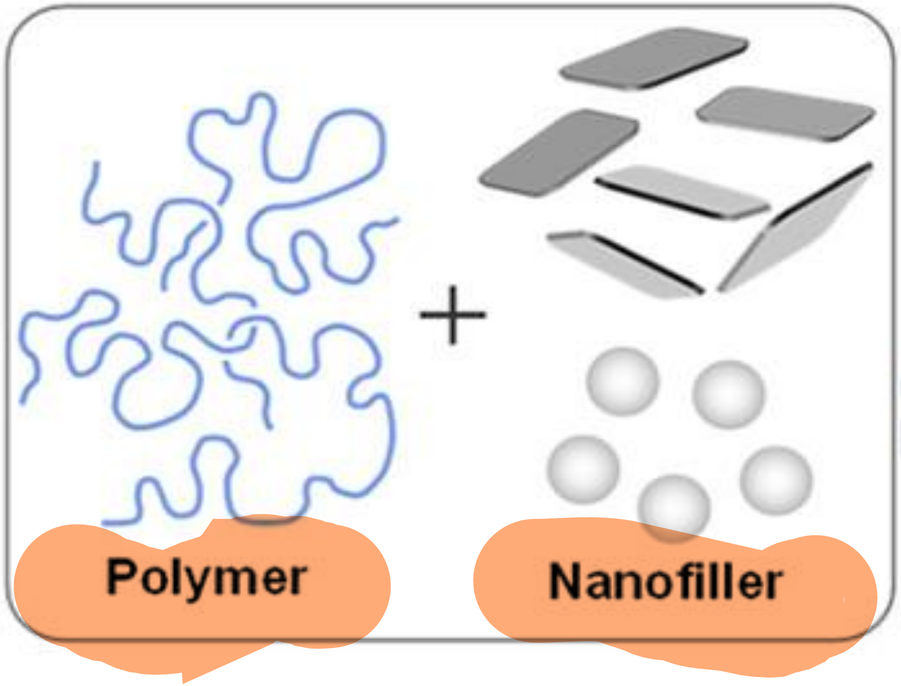
Composite materials will be having two important components Matrix and filler (reinforcement).

- Polymer composites have good mechanical, friction, durability, and wear performance after being reinforced with waste or nanofiller.

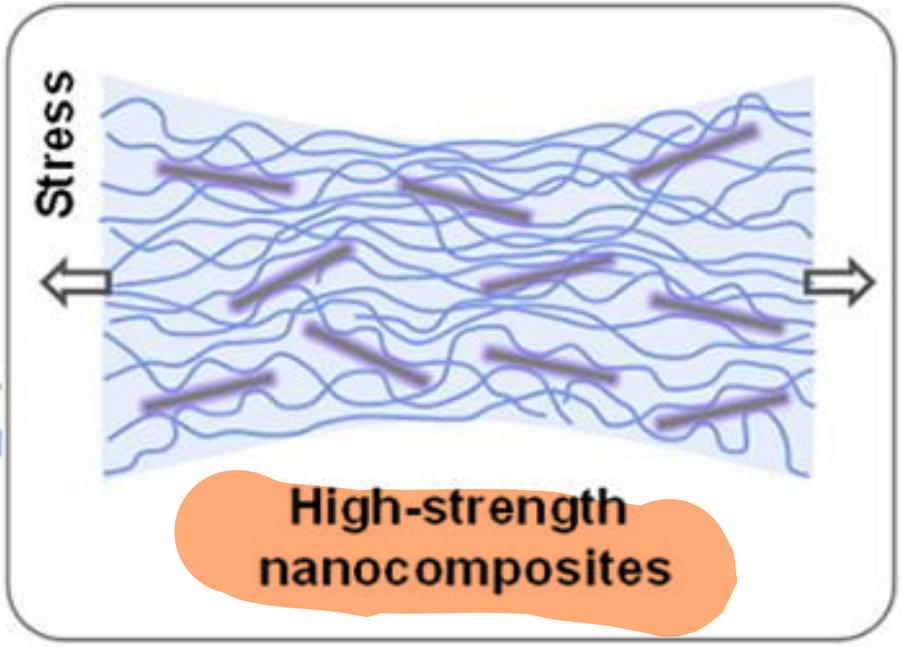


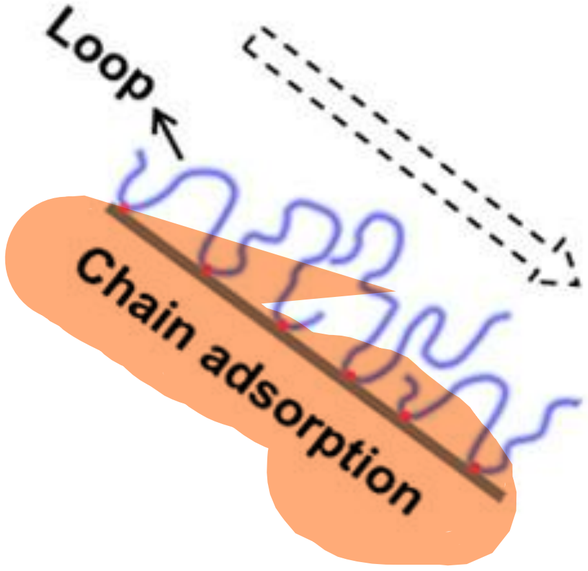


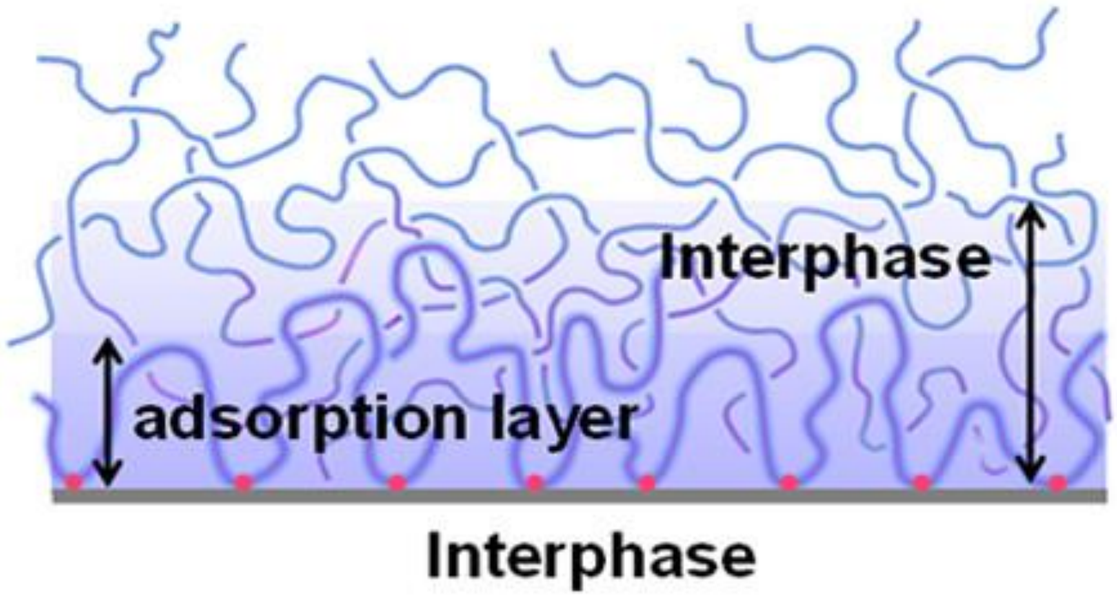
Polymer Composites

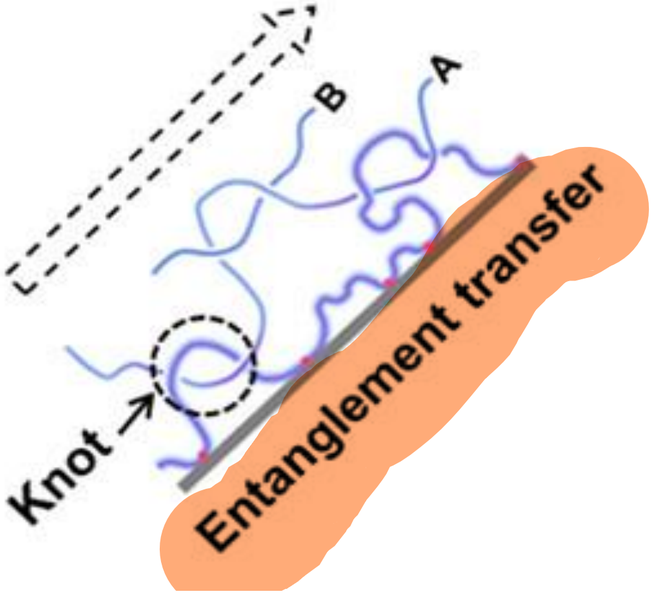












Glass fiber

Most versatile industrial materials known today.

Produced from raw materials, which are available in unlimited supply

They exhibit useful bulk properties such as

- Hardness
- Resistance to chemical attack
- Stability and
- Inertness, as well as desirable



Glass fiber

Glass fibers are manufactured from molten glass, from which glass monofilaments are drawn and then gathered to strands.

The strands are used for preparation of different glass fiber products (yarns, rovings, woven fabrics, mats).

The most popular matrix materials for manufacturing fiberglasses are Thermosets such as unsaturated polyesters (UP), epoxies (EP) and Thermoplastics such as nylon (polyamide), polycarbonate (PC), polystyrene (PS), polyvinylchloride (PVC).

Fiber glass materials usually have laminate structure with different fibers orientations in the reinforcing glass layers. Various glass fibers orientations result in anisotropy of the material properties in the plane parallel to the laminates.

Concentration of glass fibers in fiberglass is normally about 40% - 70%.

Glass fiber

Step 1: Raw Materials Preparation

Step 2: Melting the Raw Materials

Step 3: Forming the Fiber

Step 4: Cooling and Solidification

Step 5: Sizing Application

Step 6: Formation of a Fiber Mat

Step 7: Heat Treatment (Annealing)

Step 8: Cutting and Packaging

Step 9: Quality Control

Glass fiber

Raw Materials Silica sand (SiO_2), limestone ($CaCO_3$), soda ash (Na_2CO_3)

Melting the Raw Materials:

The raw materials are heated in a furnace to a high temperature (around 1700-2000°C or 3092-3632°F) to form a molten liquid called "**glass melt**."

Forming the Fiber:

The molten glass is then forced through tiny openings (spinnerets) to create thin strands or filaments of glass. This process is known as "**fiberization.**"

Cooling and Solidification:

As the glass fibers exit the spinnerets, they rapidly cool and solidify into long, thin strands. The cooling process can involve air or water.

Sizing Application:

The glass fibers are coated with a sizing material (a chemical compound) to improve their handling, adhesion properties, and compatibility with specific applications.

sizing material : epoxy, polyester, vinyl ester, or polyurethane.

Glass fiber

Formation of a Fiber Mat:

The individual glass fibers are collected and pulled together to form a mat or strand of continuous fibers. The mat can be further processed into various forms, such as rovings, chopped strands, or mats.

Heat Treatment (Annealing):

The formed fibers are subjected to a controlled heating and annealing process to relieve internal stresses and enhance their mechanical properties.

Cutting and Packaging:

The continuous glass fiber strands are cut into specific lengths or dimensions depending on the intended application. The finished glass fibers are then packaged and prepared for distribution.

Quality Control:

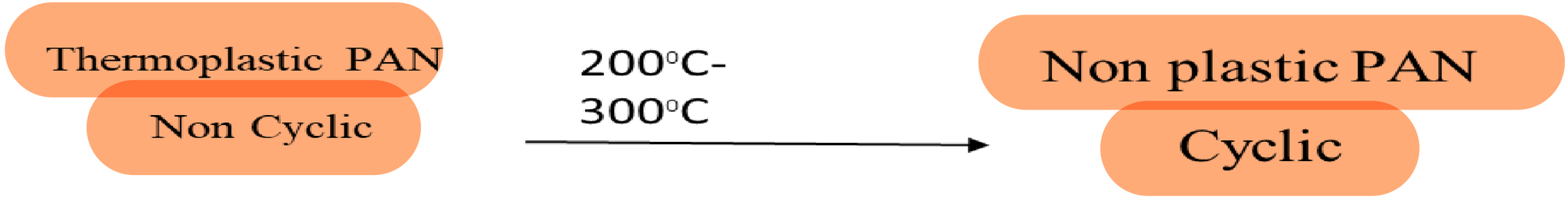
Throughout the entire manufacturing process, quality control measures are implemented to ensure the produced glass fibers meet required specifications for strength, diameter, and other properties.

CARBON FIBRES

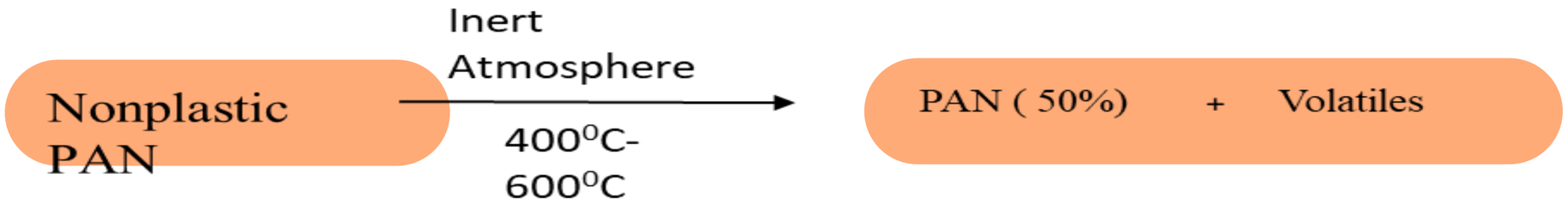
- Carbon fibers are novel high-performance materials.
- They could be described as fibers containing at least 90 % carbon obtained by the controlled pyrolysis of appropriate fibers.
- Carbon atoms are bonded together in a crystal that are aligned parallel to the axis of fibre.
- This alignment gives fibre high length to volume ratio.
- Several thousand carbon fibres are bundled together which are then woven into fabric.

Synthesis

Step-1 Oxidative stabilization- Stretching & Oxidation



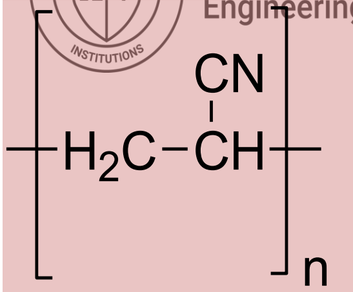
Step-II- Carbonization



Step-III-Graphatization (-N_2)

The fibres are treated at temperature between 600°C- 1300°C to improve order and orientation of crystallites in direction of fiber axis.

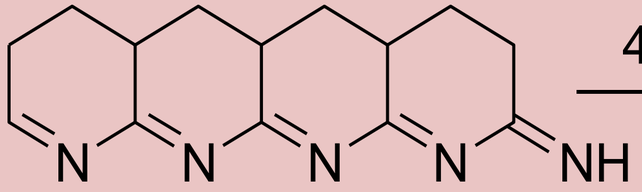
Chemical Reaction



PAN

200-300C

Oxidative stabilisantion

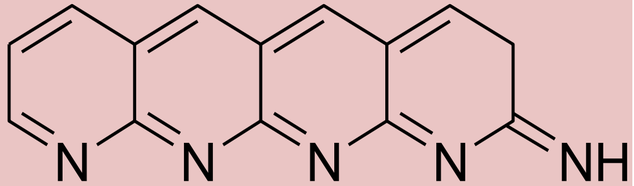


Cyclic PAN

400-600C

$-H_2$

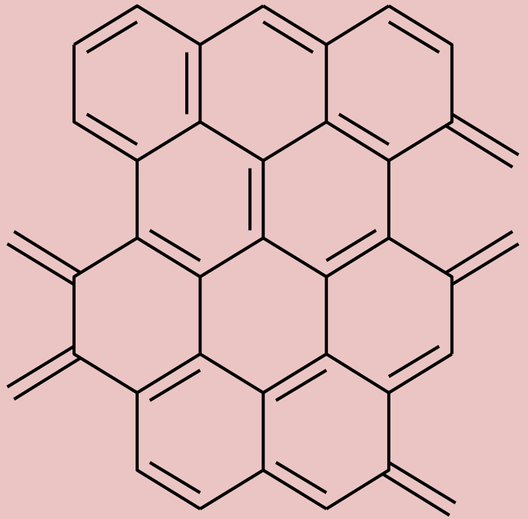
Carbonisation



Graphatisation

600-1300C

$-N_2$



Properties of Carbon fibre

- Carbon fiber has tensile strength 3 times greater than steel & 4.5 times lighter.
- Exceptional Impact properties- Good energy dissipation mechanism
- Low coefficient of thermal expansion compared to metals
- Better corrosion resistance
- High chemical inertness

Application of carbon fibres in composites

Applications in Sports:

- Hockey Sticks
- Track Spikes
- Bicycle Frames
- Helmets
- Motor Racing
- Tennis Rackets
- Golf Clubs
- Cricket Bats
- Gliders
- Surfboards
- Rowing Shells

Application of carbon fibres in composites

Application in the Automotive Industry:

- Racing car chassis
- Hoods
- Car emblems
- Mufflers
- Interior panels of a car
- Steering wheels

Application in Civil Engineering

- Carbon Fiber Reinforced Polymer is can be applied to reinforce concrete structures.

Applications in Aerospace Engineering

- Aircraft: main wings, Tail units, Rudders, Elevators, Floor panel, Beams, Lavatory units, Seats.
- Rockets: Nozzle cones, Motor cases.
- Satellites: Antennas, Solar battery panels.